

**Sales Performance and Product Demand Analysis for Improved Inventory
Planning in a Retail Garments Store**

A Mid-term report for theBDM capstone Project

Submitted by :

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1 Executive Summary

Matching Point is a retail garment and matching accessories store located at Powerhouse Road, Maripur, Muzaffarpur, Bihar. Established in 2014, the business sells sarees, suit pieces, dupattas, blouse materials, and other matching accessories. The store currently maintains sales and inventory records manually, leading to two major operational challenges: locked working capital due to slow-moving inventory and high inventory holding costs caused by overstocking and stock imbalance.

Primary data was collected directly from the business through sales registers, bill books, and inventory records for the period January 2026 to April 2026. The dataset comprises 699 sales transactions across 56 products, organized into four worksheets: Item Master, Daily Sales Data, Inventory Summary, and Current Stock. Key variables include Quantity Sold, Selling Price, Cost Price, Opening Stock, and Closing Stock, which together support analysis of product demand, revenue, and inventory movement. Descriptive statistics show an average of 3.04 units sold per transaction and an average selling price of ₹417.45, indicating moderate transaction volume with notable price variation across product categories.

The analysis applied Descriptive Statistics, Pivot Table Analysis, and Inventory Status Classification to examine sales performance and stock behaviour. Preliminary findings show Dupatta as the highest-selling product by quantity, while Suit Piece generates the highest revenue. Inventory analysis reveals 50% of products at normal stock levels, 42.9% near stockout, and 7.1% overstocked. These insights will guide the ABC Analysis, Inventory Turnover Ratio, and Moving Average Forecasting planned for the final phase.

2 Proof of Originality

1. Business Name: Matching Point
2. Type of Business: Retail Garment and Matching Accessories Store (Sole Proprietorship)
3. Owner: Md. Tarique Anwar
4. Location: Powerhouse Road, Maripur, Muzaffarpur, Bihar
5. Business Established: 19 May 2014
6. Business Hours: 10:00 AM – 10:00 PM
7. Contact Number: +91 8863950923

8. Evidence folder : [Drive link](#) (Contains authorization letter, photographs, interview video, bill book images, and the primary dataset.)
9. Authorization Letter: Attached on the organization's letterhead with signature. [Click Here](#)
10. Primary Dataset: Microsoft Excel Dataset containing : [Click Here](#)
11. Transcription of Interview Video: [Click Here](#)



Figure 1. Front View of Matching Point Store



Figure 2. InnerView of Matching Point



Figure 3. Owner of the shop

S.No.	PARTICULARS	Rate	AMOUNT
1	Shirts	100	500
2	Shirts	100	100
3	Shirts	100	100
4	Shirts	100	100
5	Shirts	100	100
6	Shirts	100	100
7	Shirts	100	100
8	Shirts	100	100
9	Shirts	100	100
10	Shirts	100	100
11	Shirts	100	100
12	Shirts	100	100
13	Shirts	100	100
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96	Shirts	100	100
97	Shirts	100	100
98	Shirts	100	100
99	Shirts	100	100
100	Shirts	100	100

Figure 4. Sample Sales Register Used for Data Collection

3 Metadata and Descriptive Statistics

3.1 Metadata

Metadata refers to the information that describes the collected dataset. It provides details about the variables, structure, and time period of the data used in the study. In this project, primary data was collected from Matching Point through sales registers, bill books, and inventory records. The collected data was organized into Microsoft Excel to support the analysis of sales performance and inventory management.

Excel data :  [Click Here](#)

Dataset Overview

Dataset Name	Worksheets	Time Period	Products	Sales Records
Sales & Inventory Dataset	4	Jan 2026 – Apei 2026	56	699

The dataset contains the following important variables:

- **Date:** Used to analyse daily and monthly sales trends.
- **Item ID:** Uniquely identifies each product.
- **Item Name:** Used for product-wise sales analysis.
- **Category:** Helps compare sales across product categories.
- **Quantity Sold:** Measures product demand.
- **Selling Price:** Used to calculate revenue.
- **Cost Price:** Used to estimate inventory investment, calculate inventory value, and support future ABC Analysis and Inventory Turnover Ratio.
- **Total Sales:** Indicates revenue generated from each transaction.
- **Opening Stock:** Shows stock available before sales.
- **Closing Stock:** Helps evaluate inventory movement.
- **Inventory Status:** Identifies overstocked, normal, and near-stockout products.

These variables were selected because they directly support the objectives of this study. Sales variables such as Quantity Sold and Total Sales help analyse product demand and revenue, while inventory variables such as Opening Stock, Closing Stock, and Inventory Status help evaluate inventory movement and stock availability. Together, these variables support the analysis of working capital utilization and inventory cost optimization.

3.2 Descriptive Statistics

Descriptive statistics are used to summarize the main characteristics of the collected dataset. They provide a clear understanding of sales performance by presenting numerical measures such as mean, median, mode, minimum, maximum, and standard deviation. These measures help identify sales patterns, product demand, pricing behaviour, and variation in sales transactions. The results provide an initial understanding of the business and support further analysis related to inventory management and working capital optimization.

	A	B	C	D	E	F	G	H
1	Date	Item ID	Item Name	Category	Sub Category	Quantity Sold	Selling Price	Total Sales
2	01-01-2024	0018	Kurti	Matching Material	Semi Rubyja	1	120	120
3	01-01-2024	0011	Suit Piece	Suit Piece	Printed Cotton	3	900	2700
4	01-01-2024	0016	Saree	Saree	Printed Saree	3	950	2850
5	01-01-2024	0017	Sareja Cloth	Matching Material	Poplin	1	65	65
6	01-01-2024	0003	Bedsheet	Home Textile	King Size (100x120)	4	1200	4800
7	01-01-2024	0021	Dupatta	Dupatta	Cotton Printed	2	200	400
8	01-01-2024	0020	Dupatta	Dupatta	Cotton Plain	5	180	900
9	01-01-2024	0019	Dupatta	Dupatta	Cotton Printed	2	150	300
10	01-01-2024	0023	Dupatta	Dupatta	Cotton Printed	4	350	1400
11	01-01-2024	0018	Kurti	Matching Material	Semi Rubyja	5	120	600
12	01-01-2024	0058	Masi	Night Wear	Cotton Masi	3	500	1500
13	01-01-2024	0007	Kurti	Ladies Wear	Cotton-Kurti	4	950	3800
14	01-01-2024	0017	Sareja Cloth	Matching Material	Poplin	2	45	90
15	01-01-2024	0029	Blouse Piece	Blouse Material	Cotton	3	180	540

Figure 5. Snapshot of the Primary Sales Dataset

Statistics	Quantity Sold	Selling price	Total sales	Opening stock	Closing stock	Cost Price
Count	699	699	699	56	56	56
Mean	3.04	417.45	1,298.38	47.45	8.66	301.43
Median	3.00	250.00	720.00	25.00	7.00	220.00
Mode	5	120	360	12	3	120
Standard Deviation	1.45	395.25	1,484.25	46.45	9.01	283.37
Minimum	1	40	40	9	0	20
Maximum	7	2,000	10,000	201	53	1500

Table 4.1 Descriptive Statistics of the Primary Dataset

The descriptive statistics provide a summary of the collected sales dataset. The average quantity sold per transaction is 3.04 units, while the average selling price is ₹417.45. The mean total sales value is ₹1,298.38. The standard deviation indicates variation in selling price and total sales across different products. Opening Stock averages 47.45 units per product with high variation (SD=46.45), while Closing Stock averages only 8.66 units, indicating limited buffer stock across most products. The average Cost Price of ₹301.43 reflects a wide product price range, supporting the need for value-based ABC classification in the final phase.

The minimum and maximum values show that the business deals with products of different price ranges and sales volumes. These statistics provide a basic understanding of sales behaviour and support further inventory and sales analysis. In addition to sales variables, inventory-related variables such as Opening Stock, Closing Stock, and Cost Price were also prepared for further inventory analysis. These variables will support inventory optimization, ABC Analysis, and Inventory Turnover Analysis during the final phase of the project.

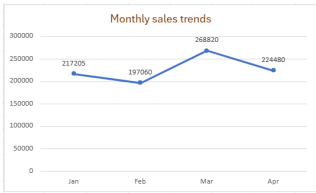


Figure 6. Monthly Sales Trend

Figure 6 shows the monthly sales trend of Matching Point from January to April 2026. Sales decreased slightly from ₹217,205 in January to ₹197,060 in February. A significant increase was observed in March (₹268,820) due to higher customer demand during the festive and wedding season. Sales declined slightly in April (₹224,480) but remained higher than February. This trend indicates seasonal demand, suggesting that the business should maintain higher inventory levels before the festive season to avoid stock shortages.

4 Detailed Explanation of Analysis Process / Method

4.1 Data Cleaning and Preprocessing

The collected primary data was reviewed before analysis to improve its quality and consistency. The following data cleaning and preprocessing steps were performed:

- Missing values were checked, and no important data was found to be missing.
- Duplicate records were identified and removed to avoid repeated sales entries.
- Product names, categories, and sub-categories were standardized to maintain consistency.

Opening Stock, Quantity Sold, and Closing Stock values were verified for logical correctness.

- The Total Sales column was calculated using the formula:

$$\text{Total Sales} = \text{Quantity Sold} \times \text{Selling Price}$$

- Inventory status was verified using the Closing Stock values and classified into Normal, Near Stockout, and Overstock.
- After cleaning, the dataset was organized in Microsoft Excel for further analysis using Pivot Tables and descriptive statistics.

Importance of Data Cleaning

Data cleaning improves the accuracy, reliability, and consistency of the dataset. It removes errors and duplicate records, making the analysis more reliable. Clean data helps generate meaningful business insights and supports better inventory management and sales analysis.

Raw Data Organization:

The raw data was collected from the business's handwritten sales registers and customer bills. The collected information was manually organized in Microsoft Excel into separate worksheets such as Daily Sales Data etc. During the interview, the owner highlighted that the growing shift of customers towards online shopping has become one of the major challenges for the business.

4.2 Descriptive Statistical Analysis

Methods:

Descriptive statistical analysis was performed using Microsoft Excel to summarize the sales dataset collected from January 2026 to April 2026. Measures such as count, mean, median, mode, minimum, maximum, and standard deviation were calculated for Quantity Sold, Selling Price, and Total Sales. Metadata describing the dataset, variables, and business information was also prepared to understand overall sales behaviour.

Justification:

Descriptive statistics provide a clear understanding of sales distribution, customer purchasing behaviour, and product performance. These measures establish the baseline for further inventory and sales analysis while identifying variations in transaction values across different products.

4.3 ABC Analysis

Methods:

ABC Analysis has been identified as one of the analytical techniques that will be used in this project to classify products based on their cumulative revenue contribution. The preliminary sales dataset has been arranged in descending order of sales value to prepare the data for ABC classification. The detailed classification of products into Category A, B, and C and the corresponding Pareto-based analysis will be carried out in the final phase of the project.

Justification:

The preliminary revenue distribution provides an initial understanding of product sales concentration and helps identify products that contribute significantly to business revenue. A complete ABC classification along with managerial recommendations will be presented in the final report after completing the detailed cumulative percentage analysis.

4.4 Inventory Status Analysis**Methods:**

Inventory records were analysed by comparing available stock with estimated demand. Products were classified into Overstock, Normal Stock, and Near Stockout categories using inventory quantity available at the end of the study period.

Justification:

Inventory status analysis identifies products requiring replenishment as well as products carrying excess stock. This helps minimize inventory holding costs while avoiding stock shortages.

4.5 Inventory Turnover Analysis**Methods**

Inventory Turnover Analysis will be performed by comparing product sales with average inventory available during the study period. Products with low turnover indicate slow-moving inventory, while products with high turnover indicate efficient inventory utilization.

Justification

This analysis helps identify products responsible for locked working capital and supports better inventory planning by reducing excess inventory while preventing stock shortages.

4.6 Moving Average Forecasting**Methods**

Moving Average Forecasting will be performed using monthly sales data collected from January to April 2026 to estimate future product demand.

Justification

Forecasting supports better purchasing decisions, minimizes stock shortages, and reduces unnecessary inventory holding costs.

5 Results and Findings

The collected primary data was analysed using descriptive statistics, Pivot Tables, and charts. The analysis identified important sales and inventory patterns that can help improve business performance.

The main findings are presented below ABC Analysis, Inventory Turnover Ratio, and Moving Average Forecasting will be completed during the final phase of the project using the prepared inventory dataset. The current report presents the preliminary analysis based on available sales and inventory data.

5.1 Results for problem statement 1

Reducing Locked Working Capital through Inventory Analysis

Objective

To identify products and inventory categories where business capital is unnecessarily blocked because of slow-moving inventory.



Figure 8. Revenue by Product Category

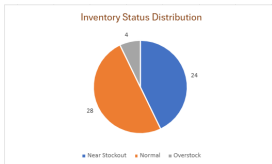


Figure 9. Inventory Status Distribution

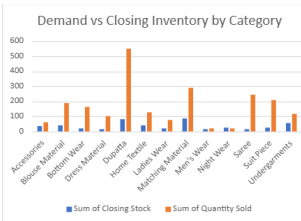


Figure 10. Demand vs Closing inventory

Interpretation

Category-wise revenue analysis shows that Suit Piece, Saree, and Dupatta generate the highest revenue and should receive higher inventory priority. Inventory Status Distribution indicates that 42.9% of products are close to stockout while only 7.1% are overstocked. The comparison between Quantity Sold and Closing Stock identifies categories where inventory investment is higher than demand. These findings indicate that capital is currently blocked in slow-moving inventory while fast-moving products require timely replenishment. Better inventory prioritization using ABC Analysis can significantly improve working capital utilization.

5.2 Results for problem statement 2

Minimizing Inventory Holding Cost through Demand-Based Inventory Planning



Figure 11. Top selling product

5.3 Monthly Sales Trend



Figure 12. Monthly Sales Trend (January–April 2026)

Interpretation

Dupatta, Suit Piece, and Saree show consistently high customer demand throughout the study period. Monthly sales increased significantly during March, indicating seasonal demand before the festive period. Maintaining higher inventory for high-demand products while reducing stock levels of slow-moving products can reduce inventory holding costs without affecting product availability. Demand forecasting and inventory turnover analysis can further optimize purchasing decisions.

5.4 Key Findings

Based on the analysis, the following business insights were identified:

- Dupatta is the highest-selling product in terms of quantity sold.
- Suit Piece generates the highest revenue among all product categories.
- March recorded the highest monthly sales due to seasonal demand.
- Half of the products are maintained at normal stock levels.
- Nearly 43% of products are close to stockout and require timely replenishment.
- A small number of products are overstocked and should be monitored to reduce inventory holding costs.
- The analysis supports better inventory planning, product stocking decisions, and working capital management.
- ABC Analysis will be performed in the final report to classify products based on cumulative revenue contribution and support inventory prioritization.
- The business owner reported that increasing customer preference for online shopping has reduced offline customer footfall. The business is therefore planning to expand its presence through online sales channels in the future.